Mobile Developer Technical Assessment

Mobile & AI Application Developer | DB Investing | Candidate brief

Decisions

Why this stack:

To achieve this I went with a cross-platform app, not separate native apps, because the brief asks for one codebase and recommends Expo. I chose React Native and TypeScript over Flutter and Dart because it is easier to set up and test on a simulator or emulator, with no paid Apple account.

Auth and Storage approach:

For auth and storage I used Supabase email and password, so sign-up, login, logout, and a session that returns after a restart are already handled. PocketBase, a small Node or Python API, or accounts stored only on the phone would have meant another server or writing auth myself. The token is saved with expo-secure-store in the iOS Keychain and the Android Keystore, in chunks when it is too big for one item. Name, risk profile, and watchlist live on the Supabase user. The service-role key stays in the dashboard, the app only uses the public anon key, and .env is not committed.

API and signal logic:

For prices I used Binance's public hourly candles, which need no key. I left CoinGecko because it needs extra history calls, Frankfurter because it is FX, and Finnhub and Alpha Vantage because they need a key by email. The signal is a 10-hour average against a 20-hour average. I chose that crossover over RSI because it only needs the closes and the reason fits one line. The risk profile is the gap before BUY or SELL: 0.5% for conservative, 0.15% for balanced, and none for aggressive. Inside the band it stays NEUTRAL.

Skipped or would improve:

I skipped charts, server push, and a saved copy of the last signals. I would store that list in secure storage, so a failed call can still show the previous result. I used Cursor to draft the Read Me, Test for bugs, average functions, run build and build errors. I kept the rule in its own file and rejected a generated backend and a chart library, because both were outside the time I had.